Savitribai Phule Pune University Second Year of Computer Science and Engineering - 2024 Pattern		
PCC-251- CSE : Database Management Systems		
Teaching Scheme	**Credits**	**Examination Scheme**
Theory : 03 Hours/Week	03	**CCE** : 30 Marks **End-Semester:** 70 Marks

Prerequisite Courses :

1. Discrete Mathematics, Data Structures and Algorithms

Course Objectives: The course aims to:

1. To understand database concepts, design principles, and ER/EER modeling.
2. To develop SQL and PL/SQL skills for efficient database operations and procedural programming.
3. To apply normalization techniques for designing well-structured relational databases.
4. To explore database transactions, concurrency control methods, and recovery mechanisms.
5. To analyse NoSQL database models and their role in managing unstructured data.

Course Outcomes: Upon successful completion of this course, students will be able to:

- CO1: **Explain** the fundamentals of database management systems, including data models, ER modeling, and database design.
- CO2: **Develop** and **execute** SQL and PL/SQL programs to manage and manipulate relational data.
- CO3: **Apply** normalization techniques to improve database design and ensure data integrity.
- CO4: **Analyze** transaction management concepts and concurrency control techniques for reliable database systems
- CO5: **Evaluate** NoSQL database types and **explain** their suitability for handling unstructured data.

Course Contents

Unit I - Introduction to Database Management System (09 Hours)

Introduction to Database Management Systems, Purpose of Database Systems, Database-System Applications, View of Data, Database Languages, Database System Structure, Enterprise Constraints Data Models, Database Design and ER Model: Entity, Attributes, Relationships, Constraints, Keys, Design Process, Entity Relationship Model, ER Diagram, Design Issues, Extended E-R Features, Converting E-R & EER diagram into tables.

Case Study: Study of Architecture of any DBMS like Oracle or MySQL. Design a database schema for any problem given in previous Question Papers.

Unit II - SQL and PL/SQL (09 Hours)

SQL: DDL, DML, Select Queries, String, Date and Numerical Functions, Aggregate Functions ,View, Indexes, Group by and Having Clause, Join Queries, Set, Set operation, Set membership, Nested queries, DCL, TCL
PL/SQL: Control Statement, Cursor, Stored Procedure and Function, Trigger

Case Study : Design and implement a Student Course Management System using SQL and PL/SQL to manage students, courses, and faculty members efficiently. The system should store and retrieve relevant data, ensuring integrity, security, and performance optimization.

Unit III - Relational Database Design (09 Hours)

Relational Model: Basic concepts, Attributes and Domains, CODD's Rules, Relational Integrity, Referential Integrities, Database Design: Features of Good Relational Designs, Normalization, Atomic Domains and First Normal Form, Decomposition using Functional Dependencies, 2NF, 3NF, BCNF.

Case study: Design and Optimization of a Relational Database for a University Management System

Unit IV - Database Transactions (09 Hours)

Basic concept of a Transaction, Transaction Management, Properties of Transactions, ACID, Concept of Schedule, Serial Schedule, Serializability: Conflict and View, Cascaded Aborts, Recoverable and Non-recoverable Schedules, Concurrency Control: Need, Locking Methods.

Case study : Design Online Shopping Cart Transaction Management In an e-commerce platform, multiple users simultaneously add, update, and purchase products. To ensure data consistency and reliability, the system must handle concurrent transactions effectively.

Unit V - NoSQL Database (09 Hours)

Introduction to NoSQL Database, NoSQL data models, CAP theorem and BASE Properties, Comparative study of SQL and NoSQL, MongoDB: CRUD Operations, Indexing and Aggregation.

Case study: Study NoSQL Database Selection for a Social Media Platform.

Learning Resources

Text Books:

1. Silberschatz A., Korth H., Sudarshan S., "Database System Concepts", McGraw Hill Publishers, ISBN 0-07-120413-X, 6th edition
2. Connally T., Begg C., "Database Systems", 4th Edition, Pearson Education, 2002, ISBN 8178088614
3. D T Editorial Services "BIG DATA Black Book", Dreamtech Press ISBN 13 : 9789351199311

Reference Books:

1. C J Date, "An Introduction to Database Systems", Addison-Wesley, ISBN: 0201144719
2. S.K.Singh, "Database Systems: Concepts, Design and Application", Pearson Education, ISBN 978-81-317-6092-5
3. Kristina Chodorow, Michael Dierolf, "MongoDB: The Definitive Guide", O Reilly Publications, ISBN: 978-1-449-34468-9
4. Adam Fowler, "NoSQL For Dummies", John Wiley & Sons, ISBN-1118905628
5. Kevin Roebuck, "Storing and Managing Big Data - NoSQL, HADOOP and More", Emereopty Limited, ISBN: 1743045743, 9781743045749
6. Joy A. Kreibich, "Using SQLite", O'REILLY, ISBN: 13:978-93-5110-934-1
7. Ivan Bayross, "SQL, PL/SQL the Programming Language of Oracle", BPB Publications ISBN: 9788176569644, 9788176569644

MOOC / NPTEL/YouTube Links: -

1. https://nptel.ac.in/courses/106106220
2. https://nptel.ac.in/courses/106105175
3. https://www.mongodb.com/resources/basics/databases/nosql-explained
4. https://learn.microsoft.com/en-us/azure/cosmos-db/nosql/modeling-data